MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal choice.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The length of an arc of a sector of angle θ (in degrees) of a circle with radius r is: [1]

(A) $\frac{\theta}{180^\circ} \times 2\pi r$ (B) $\frac{\theta}{360^\circ} \times 2\pi r$ (C) $\frac{\theta}{180^\circ} \times \pi r^2$ (D) $\frac{\theta}{360^\circ} \times \pi r^2$

Q2. The area of a sector of angle p° of a circle with radius R is: [1]

(A) $\frac{p}{180} \times 2\pi R$ (B) $\frac{p}{180} \times \pi R^2$ (C) $\frac{p}{360} \times 2\pi R$ (D) $\frac{p}{720} \times 2\pi R^2$

Q3. If the area of a sector of a circle bounded by an arc of length 5π cm is equal to 20π cm^2, then the radius of the circle is: [1]

(A) 12 cm (B) 16 cm (C) 8 cm (D) 10 cm

Q4. If the area of a sector of a circle is $\frac{5}{18}$ of the area of the circle, then the central angle of the sector is: [1]

(A) 60° (B) 90° (C) 100° (D) 120°

Q5. **Assertion (A):** A running track is in the form of a circular ring. If the inner and outer circumferences are 352 m and 396 m respectively, then the width of the track is 7 metres.
Reason (R): The width of a circular annulus with outer circumference C_2 and inner circumference C_1 is given by $w = \frac{C_2 - C_1}{2\pi}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. The circumference of a circle exceeds its diameter by 16.8 cm. Find the circumference of the circle. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Find the area of the shaded region in a square $ABCD$ of side 14 cm, where four semicircles are drawn with each side of the square as diameter. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

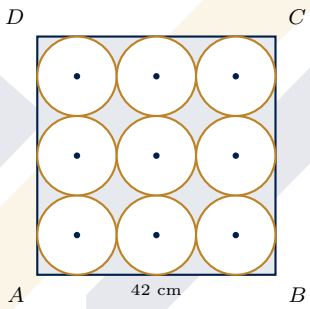
Q8. An equilateral triangle of side 12 cm has an inscribed circle touching its sides. Find the area of the region inside the triangle but outside the circle. (Use $\pi = 3.14$, $\sqrt{3} = 1.732$) [3]

Q9. Two circular flower beds are set up on two opposite sides of a square lawn $ABCD$ of side 56 m. The centre of each circular flower bed is the intersection point O of the diagonals of the lawn. Find the sum of the areas of the lawn and the flower beds. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Handicraft Block Printing on Silk Handkerchief

A textile artisan designs a traditional block print on a square silk handkerchief of side 42 cm. The pattern consists of nine identical circular motifs arranged neatly in three rows of three circles each, touching the adjacent circles and handkerchief edges. Each circle has radius $r = 7$ cm.



(i) Calculate the total area of the square silk handkerchief. [1]

(ii) Calculate the combined area occupied by all nine circular motifs. [1]

(iii) Find the remaining unprinted area of the silk handkerchief, OR if each circular motif costs 5 to print and the remaining area is dyed at 0.05 per cm^2, find the total production cost. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

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